

hidden state

demand

normal
promo*
seasonal lift*
decline*

supplier

reliable
wobbling
degraded*
defunct*

freight

slack
normal
tightening
spike*

port

clear
building
congested*
customs hold*

quality

in control
drifting
out of control*

canal

calm, watch
recovery
false alarm
disruption*

observation boundary

dashboard

point-of-sale
+ forecast
(Gaussian noise)

scorecard band,
fill, lead slip
reads one band
better than truth

rate index
+ outlook
(noisy)

berth wait
~1 → ~16 days

inspection band
accept / marginal / reject

transit counts
closure and false alarm
emit the same crash

* stress regime ~→ noisy weekly emission chains step from the seed's random stream before any action applies, so every policy faces the same shocks